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Implications for Newborn and Child Growth Classification Using INTERGROWTH-21st and WHO Child Growth Standards

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ABSTRACT

Objective: The aim of this study was to assess how the use of the WHO Child Growth Standards (WHO GS) and the INTERGROWTH-21st Standards (IG-21st) affect the classification and interpretation of growth trajectories.

Design: A secondary data analysis of an observational cohort study.

Setting: Five health facilities participating in the Low Birthweight Infant Feeding Exploration (LIFE) study in Malawi and Tanzania.

Population or Sample: 608 moderately low birthweight (1.5 to < 2.5 kg) infants.

Methods: *Z*-scores for weight-for-age (WAZ), length-for-age (LAZ), and head circumference-for-age (HCAZ) were calculated at birth using both the WHO GS and the IG-21st Newborn Size Standard (IG-21st NBS). Longitudinal trajectories were assessed over three periods (birth–6 months, 6–12 months and 12–24 months) using WHO GS and IG-21st Preterm Postnatal Standards (IG-21st PPS). Between 12 and 24 months, all infants were assessed using WHO GS. Conditional growth models were constructed to evaluate deviation from mean growth patterns within GA subgroups.

Main Outcome Measures: WAZ, LAZ and HCAZ *z*-scores; prevalence of *z*-scores < −2, and absolute and conditional growth trajectory differences between WHO and IG-21st standards across three postnatal intervals.

Results: At birth, IG-21st NBS *z*-scores were higher than WHO GS values especially in lower GA subgroups, particularly for LAZ and HCAZ. The proportion of infants with < −2 *z*-scores was substantially lower with IG-21st PPS than WHO GS in preterm infants for weight (6.1% vs. 95.6%), length (14.4% vs. 89.2%) and head circumference (4.2% vs. 53.5%), respectively. From birth to 6 months, between-standard *z*-score differences (IG-21st PPS minus WHO GS) were largest in preterm infants (preterm overall: WAZ: +1.78, LAZ: +1.64, HCAZ: +1.80). From 6 to 12 months, differences persisted but were smaller (preterm overall: WAZ: +0.57, LAZ: +0.79, HCAZ: +0.54), while term infants showed minimal differences between the two standards.

Conclusions: The application of the WHO GS alone may misclassify preterm and term infants' growth status, particularly at birth and during early infancy. The use of the IG-21st standards provides GA-specific classifications that may more closely align with expected growth patterns in term and preterm infants and could support more accurate monitoring during the first year of life. Based on these implications analyses, we recommend the use of the IG-21st NBS for cross-sectional assessment at birth, the IG-21st PPS from 27 to 64 weeks postmenstrual age, and the WHO GS for all babies thereafter.

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1 | Background

Growth monitoring is an essential component of paediatric care and public health surveillance, enabling early identification of undernutrition, developmental delays and other health risks. The tools used to measure growth can heavily influence the conclusions drawn and the interventions implemented [1]. However, growth monitoring practices remain inconsistent within and across countries, with many programmes continuing to rely on national references despite the availability of two international growth standards developed for universal use [2, 3]. The WHO Child Growth Standards (WHO GS) were developed based on a population of healthy, term-born, breastfed infants across six countries. They are widely adopted in global and national programmes [4]. Yet, these standards do not account for newborn size by gestational age (GA) at birth or separate charts for preterm infants at birth and postnatally.

Even among term infants, GA at birth (e.g., early vs. late term) influences early growth, and such variation may be exaggerated or obscured depending on the growth standard used. The INTERGROWTH-21st Standards (IG-21st) sought to fill this gap by developing GA-specific newborn size standards (IG-21st NBS) [5]. In addition, separate IG-21st standards were also developed for those born preterm for assessments at birth and for tracking postnatal growth (IG-21st PPS). Despite their availability, IG-21st standards are often underused, partly due to confusion over when and how to apply them. This underutilisation is further compounded by a lack of consensus on GA correction practices, limited operational guidance on transitioning between IG-21st and WHO GS, and challenges in integrating GA-specific tools into routine health information systems. Applying inappropriate standards can lead to misclassification, either under- or overestimating attained size by GA with direct implications for clinical decisions and resource allocation [6]. These concerns are more pronounced during infancy, which is a critical period for growth determination and has implications for sustained impacts on future health, development and survival [7]. As global nutrition programmes intensify efforts to improve early childhood outcomes, the consistent and appropriate use of a common set of international growth standards is essential for effective monitoring and tracking of progress and targeted intervention prioritisation [2].

To provide evidence on the implication of choice of growth standards used, we use a subset of the data collected as part of the Low Birthweight Infant Feeding Exploration (LIFE) study from Malawi and Tanzania. We use this dataset to examine how the application of WHO GS and IG-21st influences growth classification and interpretation from birth to 24 months for a cohort of children born preterm and term. The aim of the analysis is to demonstrate the extent to which growth classification varies depending on the growth standard used. These analyses and implications form the justification for developing the Guidance for International Growth Standards (GIGS), which provide a unified framework and appropriate guidance that integrates the WHO GS and IG-21st Standards for consistent use across the continuum from 24 weeks gestation to 5 years of age. By promoting harmonised application through a suite of statistical tools in common software packages [8–12] and scenario-based guidance [13], GIGS supports greater comparability across studies and settings while advancing global child growth monitoring.

2 | Methods

2.1 | Study Population

From September 2019 to February 2022, the LIFE study followed an observational cohort of moderately LBW (1.5 to < 2.5 kg) infants recruited from 12 secondary and tertiary health facilities in India, Malawi and Tanzania to understand feeding practices and growth patterns. Here, we have only conducted a secondary analysis of select growth data collected from infants recruited from five health facilities in Malawi and Tanzania for illustration purposes. Additional details on the LIFE study methodology and key findings can be found in numerous publications [14–19]. Scheduled visits occurred around birth (within 72 h), weeks 1, 2, 4, 6, 10, 14, 18, 26 (6 months), 39 (9 months) and 52 (12 months) across both sites, with a final 24-month visit (104 weeks) conducted only in Tanzania.

2.2 | GA and Anthropometric Assessment

GA was determined using various sources [20]. Priority was given to the best obstetric estimate, which included the combination of first or second ultrasound and certain last menstrual period. In case these sources of GA were not available, other sources of GA indicated in the chart (i.e., fundal height) were used. Finally, in the absence of any other measures, GA was determined by the *Dubowitz* examination at birth [21]. Anthropometric measurements included weight, recumbent length and head circumference (HC), measured by trained staff using standardised protocols.

2.3 | Key Outcomes

To assess GA–specific differences in growth classification, we compared birth measurements using the WHO GS and the IG-21st NBS, following the general approach described by Perumal et al. [22, 23] Weight, length and head circumference were converted to sex-standardised *z*-scores using WHO GS and IG-21st reference distributions. These *z*-scores were then used to classify infants as underweight (WAZ < −2), stunted (LAZ < −2), or microcephalic (HCAZ < −2).

For longitudinal follow-up to 24 months, growth was assessed using three classification approaches: (1) WHO GS applied at all timepoints; (2) the GIGS-recommended approach, applying IG-21st NBS for all newborns at birth and IG-21st PPS for preterm infants until 64 weeks postmenstrual age (PMA), then WHO GS thereafter; and (3) WHO GS at all timepoints with GA correction for preterm infants. Outcomes were summarised for WAZ, LAZ and HCAZ over the intervals birth–6 months, 6–12 months and 12–24 months.

2.4 | Statistical Analysis

For cross-sectional comparisons at birth, sex-standardised *z*-scores were generated for weight, length and head circumference using WHO GS and IG-21st NBS. The prevalence of those below −2 *z*-scores for weight, length, or head circumference

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was calculated separately under each standard and compared using percentage agreement, sensitivity, specificity and Cohen's kappa.

Longitudinal analyses were conducted for three predefined age intervals: birth–6 months, 6–12 months and 12–24 months. For each interval, growth was determined and classified based on absolute and conditional growth. Absolute growth was calculated as the absolute difference in *z*-scores between timepoints, and conditional growth as the residual from a linear regression representing deviation from the average change among children with the same starting attained size in *z*-scores. The purpose of this analysis was to evaluate conditional size, accounting for baseline attained size and therefore we did not adjust for any variable as the analysis was not focusing on establishing epidemiological associations. Models were fitted separately for each outcome (i.e., WAZ, LAZ and HCAZ).

All analyses were stratified by GA category, defined as: extremely preterm ($<28^{+0}$ weeks), very preterm (28^{+0} to 31^{+6} weeks), moderately preterm (32^{+0} to 33^{+6} weeks), late preterm (34^{+0} to 36^{+6}), early term (37^{+0} to 38^{+6} weeks), full term (39^{+0} to 40^{+6} weeks), late term (41^{+0} to 42^{+6} weeks) and very late term ($>42^{+6}$ weeks). Categories with sparse observations were combined in sensitivity summaries.

All statistical analyses were performed in R v4.3.3 (R Core Team, 2023) [24]. *Z*-scores were calculated using the GIGS package [2, 8–10] and figures were produced with ggplot2 [23].

2.5 | Ethics Approval

The LIFE study received ethical approval from 11 institutional review boards in India, Malawi, Tanzania and the USA: (1) India Health Ministry's Screening Committee with Indian Council of Medical Research acting as its secretariat (2019–2674); (2) Directorate of Health and Family Welfare Services, Government of Karnataka which also covers investigators at Women and Children Hospital, Davangere and Chigateri General District Hospital, Davangere (NHM/SPM/04/2019-20); (3) Institutional Ethics Committee of KLE Academy of Higher Education and Research which also covers investigators at JN Medical College, Belagavi and KLES Dr. Prabhakar Kore Hospital & Medical Research Center, Belagavi (KAHER/IEC/2019-20/D-2760); (4) Institutional Ethics Review Board of SS Institute of Medical Sciences & Research Centre (IERB/200/2019); (5) Institutional Ethics Committee of JJM Medical College (JJMMC/IEC-01/2019) which also covers investigators at Bapuji Child Health Institute & Research Centre, Davangere, Women & Children Hospital, Davangere and Chigateri General District Hospital, Davangere; (6) Research and Ethics Committee, Directorate of Health Services, Odisha State which also covers investigators at City Hospital Oriya Bazar, Cuttack (155/PMU/187/17); (7) Institutional Ethical Committee, Sriram Chandra Bhanja Medical College, Cuttack (7188); (8) Malawi National Health Sciences Research Committee (NHSRC2019/Protocol19/03/2250-UNCPM 21905); (9) Tanzania National Institute of Medical Research (NIMR/HQ/R.8a/Vol.IX/3126); (10) Muhimbili University of Health and Allied Sciences (DA.282/298/01.C/); and (11) USA—Harvard T.H Chan School of Public Health (IRB10-0282) which also covers investigators at Boston Children's Hospital, Brigham and Women's Hospital, Emory University, PATH and University of North Carolina (Trial registration numbers: NCT04002908 and CTRI/2019/02/017475).

3 | Results

3.1 | Population Baseline Characteristics

The data used for the illustrative analysis comprised observations from 608 moderately low-birthweight infants born to 606 mothers (Table S1). Of these infants, 281 (46.2%) were male and 327 (53.8%) were female. When disaggregating small vulnerable newborn types, Malawi and Tanzania had a higher proportion of preterm births (53.8% and 66.6%, respectively) compared to term small-for-gestational-age (SGA; < 10th percentile) births.

3.2 | Cross-Sectional Analysis at Birth: WHO GS Versus IG-21st NBS

At birth, IG-21st NBS *z*-scores were generally higher than WHO GS values among preterm infants, particularly in the very, moderately and late preterm groups, with the largest differences observed for LAZ and HCAZ (Table 1). Notably, differences were observed for LAZ and HCAZ, with mean values among preterm infants approximately 2.5 SD higher when using the IG-21st NBS compared to the WHO GS. Infants born beyond 42^{+6} weeks' gestation fell outside the original IG-21st reference range; however, within the GIGS framework, extrapolation beyond this limit permits *z*-score estimation for these infants. The magnitude of difference was greatest among infants born < 34 weeks GA, with mean LAZ and HCAZ *z*-scores more than two standard deviation lower using WHO GS compared to IG-21st NBS. In contrast, *z*-scores for full-term and late-term infants were broadly similar across the two standards.

At birth, a higher proportion of infants had *z*-scores <-2 with WHO GS compared to IG-21st NBS (GIGS). Among preterm infants overall, this was 95.6% vs. 6.1% for WAZ, 89.2% vs. 14.4% for LAZ and 53.5% vs. 4.2% for HCAZ. Among term infants overall, the corresponding proportions were: 90.3% vs. 65.0% (WAZ), 77.5% vs. 57.2% (LAZ) and 35.2% vs. 17.8% (HCAZ). For full-term infants, specifically, the proportions were comparable: 93.9% vs. 92.8% (WAZ), 70.8% vs. 62.5% (LAZ) and 39.6% vs. 39.6% (HCAZ) (Table 2).

Figure 1a–c illustrates the birth WAZ, LAZ and HCAZ among children using the WHO GS compared with the IG-21st NBS.

In addition to differences in classification at the <-2 *z*-score cut-off (Figure 2), we also examined the extent of discrepancy between the two growth standards. Large differences were observed mainly among preterm infants. For very preterm, moderately preterm and late preterm infants, 100% had *z*-score discrepancies exceeding ±0.5 *z*-scores for WAZ, LAZ and HCAZ. Among early-term infants, the respective proportions

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TABLE 1 | Birth weight, length, head circumference-for-age *z*-scores using the WHO GS and GIGS-recommended growth standards (IG-21st NBS).

Gestational age category[a]		GIGS-recommended growth standard (IG-21st NBS)	WHO GS	Mean difference between GIGS and WHO *z*-scores[b]
Weight-for-age *z*-score	***n***	**Mean (SD)**	**Mean (SD)**	**Mean (SD)**
Overall (28^{+0} to 42^{+6})	599	−1.21 (1.35)	−2.96 (0.68)	1.75 (1.42)
Preterm (28^{+0} to 36^{+6})	362	−0.49 (1.2)	−3.07 (0.68)	2.58 (1.23)
Very preterm (28^{+0} to 31^{+6})	40	1.66 (1.06)	−3.46 (0.78)	5.12 (0.84)
Moderately preterm (32^{+0} to 33^{+6})	74	0.21 (0.76)	−3.3 (0.66)	3.51 (0.39)
Late preterm (34^{+0} to 36^{+6})	248	−1.05 (0.74)	−2.94 (0.63)	1.89 (0.52)
Term (37^{+0} to 42^{+6})	237	−2.31 (0.67)	−2.8 (0.65)	0.5 (0.45)
Early term (37^{+0} to 38^{+6})	168	−2.07 (0.57)	−2.79 (0.63)	0.72 (0.25)
Full term (39^{+0} to 40^{+6})	49	−2.8 (0.56)	−2.92 (0.75)	0.12 (0.24)
Late term (41^{+0} to 42^{+6})	20	−3.06 (0.39)	−2.6 (0.48)	−0.47 (0.15)
Length-for-age *z*-score		**Mean (SD)**	**Mean (SD)**	**Mean (SD)**
Overall (28^{+0} to 42^{+6})	597	−1.31 (1.26)	−3.02 (1.03)	1.71 (1.33)
Preterm (28^{+0} to 36^{+6})	361	−0.76 (1.16)	−3.24 (1)	2.49 (1.11)
Very preterm (28^{+0} to 31^{+6})	39	0.91 (0.86)	−3.9 (0.94)	4.81 (0.73)
Moderately preterm (32^{+0} to 33^{+6})	74	−0.16 (0.92)	−3.48 (0.98)	3.32 (0.45)
Late preterm (34^{+0} to 36^{+6})	248	−1.2 (0.93)	−3.07 (0.96)	1.87 (0.47)
Term (37^{+0} to 42^{+6})	236	−2.15 (0.89)	−2.67 (0.98)	0.52 (0.51)
Early term (37^{+0} to 38^{+6})	168	−2 (0.79)	−2.78 (0.88)	0.78 (0.27)
Full term (39^{+0} to 40^{+6})	48	−2.46 (1.08)	−2.57 (1.2)	0.11 (0.25)
Late term (41^{+0} to 42^{+6})	20	−2.64 (0.84)	−2.01 (0.93)	−0.64 (0.2)
Head circumference-for-age *z*-score		**Mean (SD)**	**Mean (SD)**	**Mean (SD)**
Overall (28^{+0} to 42^{+6})	597	−0.28 (1.31)	−1.97 (1.18)	1.69 (1.24)
Preterm (28^{+0} to 36^{+6})	361	0.17 (1.19)	−2.21 (1.18)	2.38 (1.1)
Very preterm (28^{+0} to 31^{+6})	39	1.84 (1.05)	−2.94 (1.19)	4.78 (0.84)
Moderately preterm (32^{+0} to 33^{+6})	74	0.3 (1.18)	−2.81 (1.3)	3.11 (0.53)
Late preterm (34^{+0} to 36^{+6})	248	−0.13 (0.97)	−1.92 (1.02)	1.79 (0.37)
Term (37^{+0} to 42^{+6})	236	−0.98 (1.16)	−1.61 (1.1)	0.63 (0.47)
Early term (37^{+0} to 38^{+6})	168	−0.79 (1.08)	−1.66 (1.05)	0.88 (0.21)
Full term (39^{+0} to 40^{+6})	48	−1.38 (1.3)	−1.61 (1.3)	0.23 (0.19)
Late term (41^{+0} to 42^{+6})	20	−1.68 (1.03)	−1.19 (0.96)	−0.49 (0.22)

[a]Gestational age categories (weeks^{+days}).
[b]Differences calculated with GIGS *z*-scores minus WHO *z*-scores. A positive difference here means GIGS *z*-scores were larger than the WHO *z*-scores.

were 70.8% (WAZ), 77.4% (LAZ) and 97% (HCAZ). In contrast, only a small proportion had discrepancies exceeding ±0.5 *z*-scores between the two standards among full-term infants i.e., 8.2% for WAZ, 8.3% for LAZ and 4.2% for HCAZ. For late-term infants, the proportions exceeding ±0.5 *z*-scores between the two standards were 30%, 90% and 40%, respectively (Figure S1a–c).

3.3 | Absolute Difference in Growth Comparisons Using WHO GS Versus IG-21st NBS

Between birth to 6 months, GIGS absolute change in WAZ was generally smaller compared to the difference in WAZ using the WHO GS. However, full-term absolute change in WAZ was most similar between GIGS and WHO GS. Similar findings were

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TABLE 2 | Prevalence of underweight (WAZ < −2), stunting (LAZ < −2) and microcephaly (HCAZ < −2) at birth using the WHO GS relative to the IG-21st NBS.

		WHO GS	GIGS	Sensitivity	Specificity
Gestational age category[a]	***n***	***n* (%)**	***n* (%)**	**% (95% CI)**	**% (95% CI)**
Weight-for-age (WAZ) < −2					
Overall (28^{+0} to 42^{+6})	599	560 (93.6)	176 (29.4)	30.2 (26.4–34.2)	82.1 (66.5–95.2)
Preterm (28^{+0} to 36^{+6})	362	346 (95.6)	22 (6.1)	6.4 (4.9–9.5)	100 (79.4–100)
Very preterm (28^{+0} to 31^{+6})	40	39 (97.5)	0 (0.0)	0 (0–0)	100 (25–100)
Moderately preterm (32^{+0} to 33^{+6})	74	72 (97.3)	0 (0.0)	0 (0–0)	100 (15.8–100)
Late preterm (34^{+0} to 36^{+6})	248	235 (94.8)	22 (8.9)	9.4 (6.3–13.8)	100 (75.3–100)
Term (37^{+0} to 42^{+6})	237	214 (90.3)	154 (65.0)	68.7 (62.7–74.8)	69.4 (47.8–86.8)
Early term (37^{+0} to 38^{+6})	168	153 (91.3)	85 (50.6)	56.8 (48.6–64.7)	100 (59.5–100)
Full term (39^{+0} to 40^{+6})	49	46 (93.9)	51 (92.8)	100 (91.4–100)	0 (0–0)
Late term (41^{+0} to 42^{+6})	20	18 (90.0)	20 (100)	100 (81.5–100)	0 (0–84.2)
Length-for-age (LAZ) < −2					
Overall (28^{+0} to 42^{+6})	597	505 (84.6)	187 (31.3)	35.6 (31.5–40.0)	92.4 (84.9–96.9)
Preterm (28^{+0} to 36^{+6})	361	322 (89.2)	52 (14.4)	16.1 (12.3–20.6)	100 (91–100)
Very preterm (28^{+0} to 31^{+6})	39	38 (97.4)	0 (0.0)	0 (0–0)	100 (25–100)
Moderately preterm (32^{+0} to 33^{+6})	74	68 (91.9)	0 (0.0)	0 (0–5.3)	100 (54.1–100)
Late preterm (34^{+0} to 36^{+6})	248	216 (87.1)	52 (21.0)	24.1 (18.5–30.3)	100 (89.1–100)
Term (37^{+0} to 42^{+6})	236	183 (77.5)	135 (57.2)	69.9 (62.7–76.5)	86.8 (74.7–94.5)
Early term (37^{+0} to 38^{+6})	168	140 (83.3)	85 (51.2)	61.4 (52.8–69.5)	100 (87.7–100)
Full term (39^{+0} to 40^{+6})	48	34 (70.8)	30 (62.5)	81.3 (64.1–92.2)	100 (75.9–100)
Late term (41^{+0} to 42^{+6})	20	9 (45.0)	16 (80.0)	100 (66.4–100)	36.4 (10.9–69.2)
Head-circumference-for-age (HCAZ) < −2					
Overall (28^{+0} to 42^{+6})	597	276 (46.2)	57 (9.5)	19.6 (15.1–24.7)	99.1 (97.3–99.8)
Preterm (28^{+0} to 36^{+6})	361	193 (53.5)	15 (4.2)	4.7 (2.2–8.7)	100 (87.8–100)
Very preterm (28^{+0} to 31^{+6})	39	30 (76.9)	0 (0.0)	0 (0–11.6)	100 (64.6–100)
Moderately preterm (32^{+0} to 33^{+6})	74	56 (75.7)	2 (2.7)	3.6 (0.4–12.3)	100 (81.5–100)
Late preterm (34^{+0} to 36^{+6})	248	107 (43.1)	7 (2.8)	6.5 (2.7–13.0)	100 (97.1–100)
Term (37^{+0} to 42^{+6})	236	83 (35.2)	42 (17.8)	34.2 (25.5–43.6)	97.8 (93.5–99.4)
Early term (37^{+0} to 38^{+6})	168	61 (36.3)	23 (13.7)	37.7 (25.6–51.6)	100 (93.5–100)
Full term (39^{+0} to 40^{+6})	48	19 (39.6)	19 (39.6)	54.2 (29.1–77.1)	100 (75.9–100)
Late term (41^{+0} to 42^{+6})	20	5 (25.0)	6 (30.0)	100 (47.8–100)	93.3 (68.1–99.8)

[a]Gestational age categories (weeks^{+days}).

found in the ΔLAZ and ΔHCAZ, with variation by GA group: under the GIGS approach (using IG-21st NBS), early-term infants had smaller changes in LAZ (−0.24 vs. +0.54) and HCAZ (−0.25 vs. +0.73) compared with the WHO GS, whereas late-term infants exhibited larger positive changes in LAZ (+1.13 vs. +0.49) and HCAZ (+0.74 vs. +0.26) (Table 3, Figure S2a–c).

At 6–12 months, ΔWAZ remained smaller under the GIGS approach than WHO GS across preterm groups: very preterm −1.24 vs. +0.07; moderately preterm −0.34 vs. +0.41; and late preterm −0.26 vs. +0.14 SD while mean *z*-scores in full term remained identical in both approaches (*z*-score = 0.09). For LAZ and HCAZ, ΔLAZ was −1.53 under GIGS vs. 0.32 for WHO GS; ΔHCAZ was −0.70 vs. +0.59 (GIGS vs. WHO), and was the same *z*-score for full term infants (i.e., HCAZ was 0.30) (Table S2, Figure S3a–c).

From 12–24 months, all infants were assessed with WHO GS only, since by this age all preterm infants had reached

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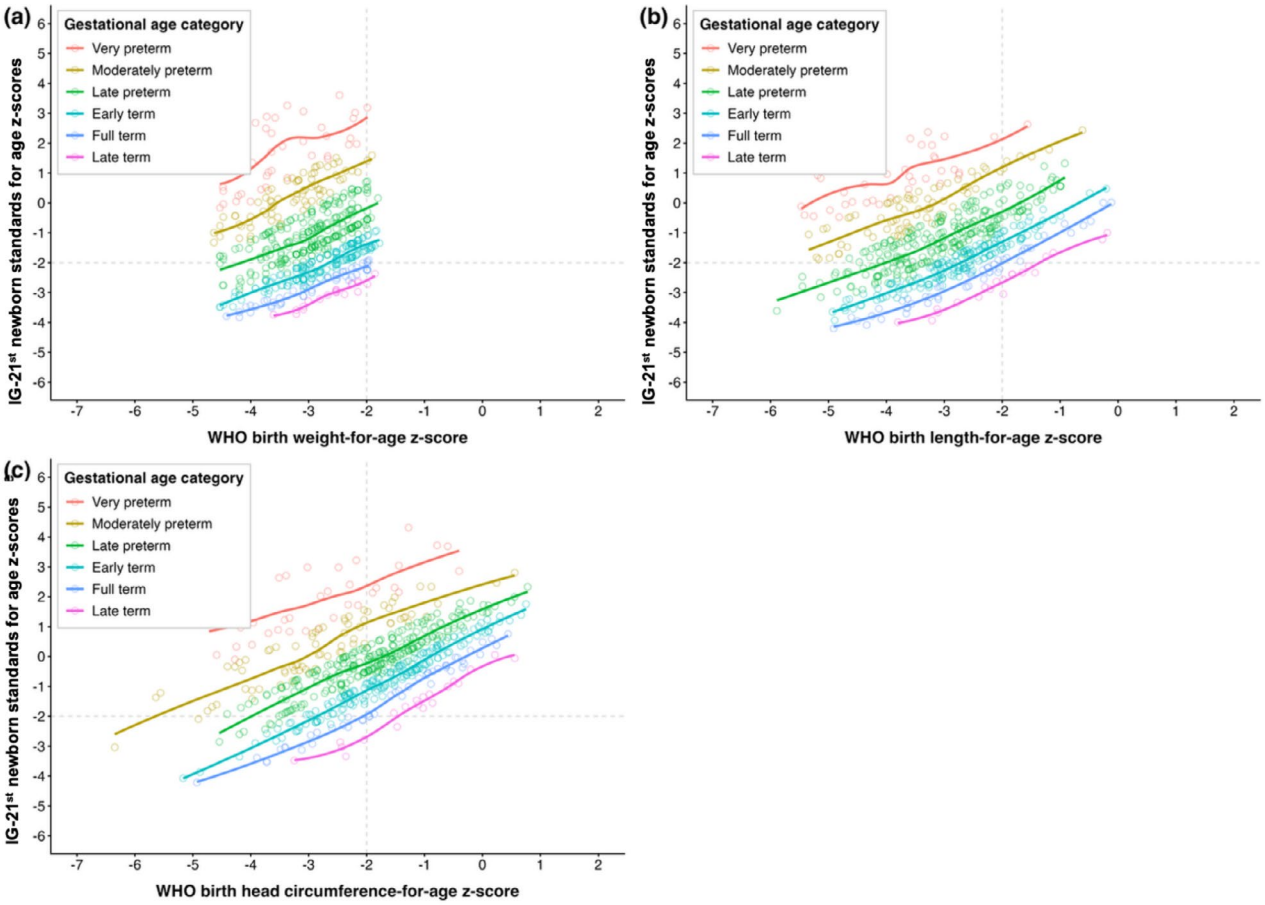

FIGURE 1 | (a) Birth weight-for-age *z*-score, (b) birth length-for-age *z*-score, (c), head circumference-for-age *z*-score among children using the WHO GS compared with the IG-21st NBS. Grey lines indicate the < −2 *z*-score cut-off for growth outcomes.

> 64 weeks PMA and both the WHO-only and GIGS approaches apply the WHO CS. WAZ decreased by 0.2–0.3 *z*-scores across most GA categories (e.g., late preterm −0.29; term −0.20), while very preterm showed a mean increase of +0.22 and full term a larger decrease of −0.53. ΔLAZ were close to zero in most groups (e.g., term −0.03), with increases in moderately preterm (+0.16) and decreases in late term (−0.46). ΔHCAZ increased overall (+0.82), with the largest increases observed in very preterm (+0.98) and moderately preterm infants (+1.18) (Table S3, Figure S4a–c).

3.4 | Conditional Difference in Growth Comparisons Using WHO GS Versus IG-21st NBS

Birth *z*-scores were positively associated with 6-month WAZ, LAZ and HCAZ under both WHO GS and IG-21st NBS (Figure S5a–c). Conditional growth from birth to 6 months differed between the two standards across GA groups (Figure 3a–c, Table S4). For WAZ, the WHO GS showed a mean conditional change of −0.04 to +0.04 *z*-scores across most preterm groups, while the IG-21st NBS showed slightly larger variability, ranging from −0.33 in very preterm to +0.18 in full term. Among term infants, the WHO GS indicated +0.20 in late term, compared with +0.71 under IG-21st, showing greater gains using the GA-specific standard. For LAZ, WHO GS values ranged from −0.10 in very preterm to +0.11 in moderately preterm, whereas IG-21st values ranged from −0.34 to +0.09 across the same GA groups. In term infants, WHO GS values were near zero (0.01 early term, −0.14 full term, +0.25 late term), while IG-21st values ranged from −0.15 to +0.92, indicating slightly higher conditional growth estimates at later gestations. For HCAZ, the WHO GS showed −0.12 in very preterm and near-zero changes in other GA groups, while IG-21st values ranged from −0.29 in very preterm to +0.32 in late term, again reflecting modestly higher conditional growth estimates using the IG-21st standard.

Conditional WAZ at 12 months was associated with 6-month WAZ (Figure S6a–c). Among preterm infants, WHO GS showed mean conditional changes from −0.19 to +0.09 *z*-scores across GA groups, while IG-21st showed −0.70 in very preterm to +0.01 in moderately preterm infants. The proportion of infants with greater than ±0.5 *z*-score differences in ΔWAZ was 90% in very preterm, 70% in moderately preterm and 40% in late preterm (Figure S7a–c, Table S5). For LAZ, WHO GS values were +0.08 in very preterm, +0.15 in moderately preterm and −0.06 in late preterm, while IG-21st values were −0.58, +0.01 and +0.1, respectively. In full-term infants, both approaches yielded similar results (−0.06 early term, +0.07 full term, +0.24 late term). For HCAZ, WHO GS showed +0.11 in very preterm and near-zero values in other groups, while IG-21st showed −0.42 in very preterm and near-zero values otherwise.

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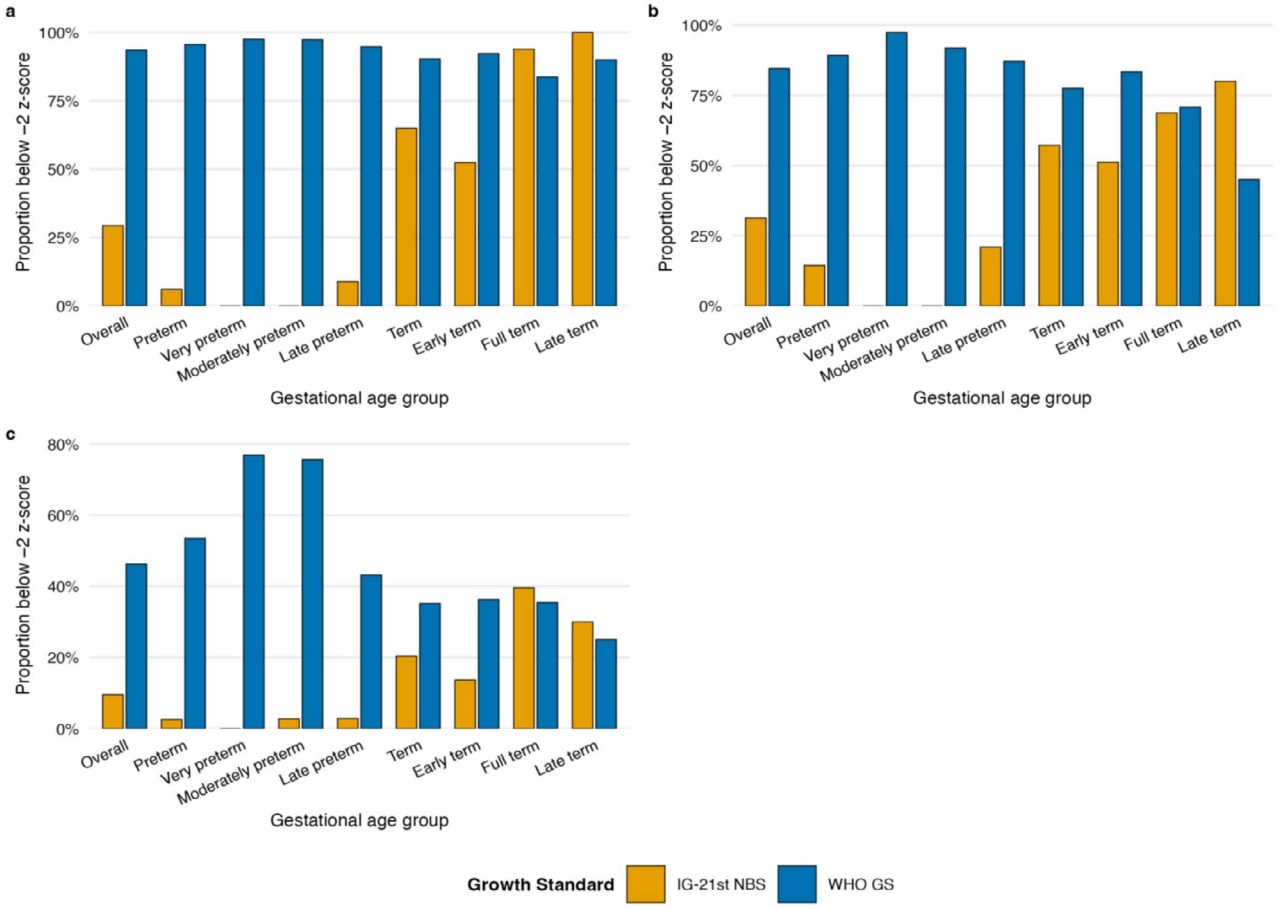

FIGURE 2 | (a) Underweight (WAZ < −2), (b) stunting (LAZ < −2) and (c) microcephaly (HCAZ < −2) prevalence at birth by gestational age category, using the WHO GS compared with the IG-21st NBS.

At 24 months, WAZ, LAZ and HCAZ were associated with their corresponding 12-month *z*-scores (correlation coefficient ranging between 0.46–0.73), similar to the 6–12 month interval (Figures S8a–c). Using WHO-GS, for WAZ, mean conditional growth was +0.26 in very preterm, +0.09 in moderately preterm, −0.07 in late preterm, and ranged from −0.11 to +0.14 among term infants. For LAZ, mean conditional growth was +0.09 in very preterm, +0.04 in moderately preterm and −0.03 in late preterm, while among term infants, mean changes ranged from −0.10 to +0.14. For HCAZ, mean changes were −0.11 in very preterm, +0.25 in moderately preterm, −0.06 in late preterm, and ranged from −0.09 to +0.09 among full term infants (Figures S9a–c, Table S6).

4 | Discussion

4.1 | Main Findings

This study builds on the foundational work of Perumal et al. (2022) [22] by replicating their methodological framework using data from a longitudinal cohort of moderately low birthweight infants in Malawi and Tanzania. Consistent with their findings in term infants, we observed that the choice of growth standard at birth, WHO GS versus IG-21st NBS, substantially influences both cross-sectional and longitudinal classification of growth, particularly in preterm and early-term infants. Our results extend their observations by including preterm infants and following growth through 24 months.

In both cross-sectional and conditional growth analyses, the WHO GS tended to overestimate catch-up growth among preterm infants, especially in the first 6 months. Differences were most pronounced in very preterm infants and narrowed with increasing GA, converging by 24 months when all infants were assessed using WHO GS. By contrast, the GIGS approach better reflected true size and growth in preterm infants, while at term it captured slower early and faster late-term growth. Conditional analyses further showed that early size strongly predicts later size, with associations strengthening after 6 months and remaining consistent through 24 months.

Our findings are consistent with the review by Ngatia et al. (2025) [3], which highlights how applying international standards versus local references often shifts classification: local charts frequently show higher weight-for-age, while WHO GS identify more children as stunted or underweight, and IG-21st tends to classify fewer SGA but more LGA. Scherdel et al. [25] similarly emphasised that growth-monitoring practices are not always well validated, and inappropriate chart choice can delay recognition of health problems. Together, these studies reinforce our observation that WHO GS can misclassify catch-up growth in preterm infants, while the structured GIGS approach reduces misclassification by clarifying which standards to appropriately apply across ages and GA groups.

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TABLE 3 | Mean changes in WAZ, LAZ and HCAZ *z*-scores at birth to 6 months.

Gestational age category	*n*	GIGS Mean (SD)	WHO Mean (SD)
Weight-for-age *z*-score			
Overall (28^{+0} to 42^{+6})	529	0.14 (1.39)	1.51 (1.13)
Preterm (28^{+0} to 36^{+6})	314	−0.11 (1.31)	1.67 (1.09)
Very preterm (28^{+0} to 31^{+6})	36	−0.23 (0.91)	1.75 (0.92)
Moderately preterm (32^{+0} to 33^{+6})	61	−0.51 (1.18)	1.77 (1.13)
Late preterm (34^{+0} to 36^{+6})	217	0.16 (1.05)	1.63 (1.11)
Term (37^{+0} to 42^{+6})	215	0.79 (1.24)	1.27 (1.14)
Early term (37^{+0} to 38^{+6})	147	0.5 (1.16)	1.22 (1.14)
Full term (39^{+0} to 40^{+6})	48	1.04 (1.19)	1.34 (1.21)
Late term (41^{+0} to 42^{+6})	20	1.89 (0.99)	1.42 (0.96)
Length-for-age *z*-score			
Overall (28^{+0} to 42^{+6})	528	−0.37 (1.3)	0.8 (1.18)
Preterm (28^{+0} to 36^{+6})	314	−0.63 (1.24)	1.01 (1.18)
Very preterm (28^{+0} to 31^{+6})	36	−1.75 (1.04)	1.17 (0.97)
Moderately preterm (32^{+0} to 33^{+6})	61	−0.95 (1.16)	1.24 (1.12)
Late preterm (34^{+0} to 36^{+6})	217	−0.31 (1.07)	0.93 (1.14)
Term (37^{+0} to 42^{+6})	214	−0.01 (1.2)	0.49 (1.11)
Early term (37^{+0} to 38^{+6})	147	−0.24 (1.18)	0.54 (1.12)
Full term (39^{+0} to 40^{+6})	47	0.2 (1.22)	0.4 (1.21)
Late term (41^{+0} to 42^{+6})	20	1.13 (0.86)	0.49 (0.85)
Head circumference-for-age *z*-score			
Overall (28^{+0} to 42^{+6})	528	−0.35 (1.35)	0.98 (1.24)
Preterm (28^{+0} to 36^{+6})	314	−0.61 (1.36)	1.19 (1.25)
Very preterm (28^{+0} to 31^{+6})	36	−1.07 (1.38)	1.49 (1.39)
Moderately preterm (32^{+0} to 33^{+6})	61	−0.94 (1.35)	1.14 (1.13)
Late preterm (34^{+0} to 36^{+6})	217	−0.24 (1.04)	0.99 (1.14)
Term (37^{+0} to 42^{+6})	214	−0.08 (1.12)	0.69 (1.18)
Early term (37^{+0} to 38^{+6})	147	−0.25 (1.13)	0.73 (1.14)
Full term (39^{+0} to 40^{+6})	47	0.13 (1.38)	0.63 (1.13)
Late term (41^{+0} to 42^{+6})	20	0.74 (0.87)	0.26 (0.92)

4.2 | Strengths and Limitations

A key strength of our analysis is the use of a multi-country cohort of moderately LBW infants in two African countries. Unlike Perumal et al. [22], who focused exclusively on term-born infants, our cohort includes the full distribution of GAs, allowing for more nuanced insights into how standard choice affects growth assessment across the GA continuum. In addition, our longitudinal follow-up through 24 months provides evidence on how early growth misclassification evolves over time.

While our analyses are robust and benefit from high follow-up coverage through infancy, there are some limitations. For example, GA in the LIFE cohort was assessed using various sources (best obstetric estimate only where available), which may in some cases be less precise than ultrasound-based GA estimation. Although our study includes preterm infants, the generalisability of findings may be limited to moderately LBW infants.

4.3 | Interpretation

Together with Perumal et al., our findings reinforce that IG-21st NBS standards are not interchangeable with WHO GS at birth, particularly for infants born early or late in gestation. In our cohort, the tendency of WHO GS to overestimate early catch-up

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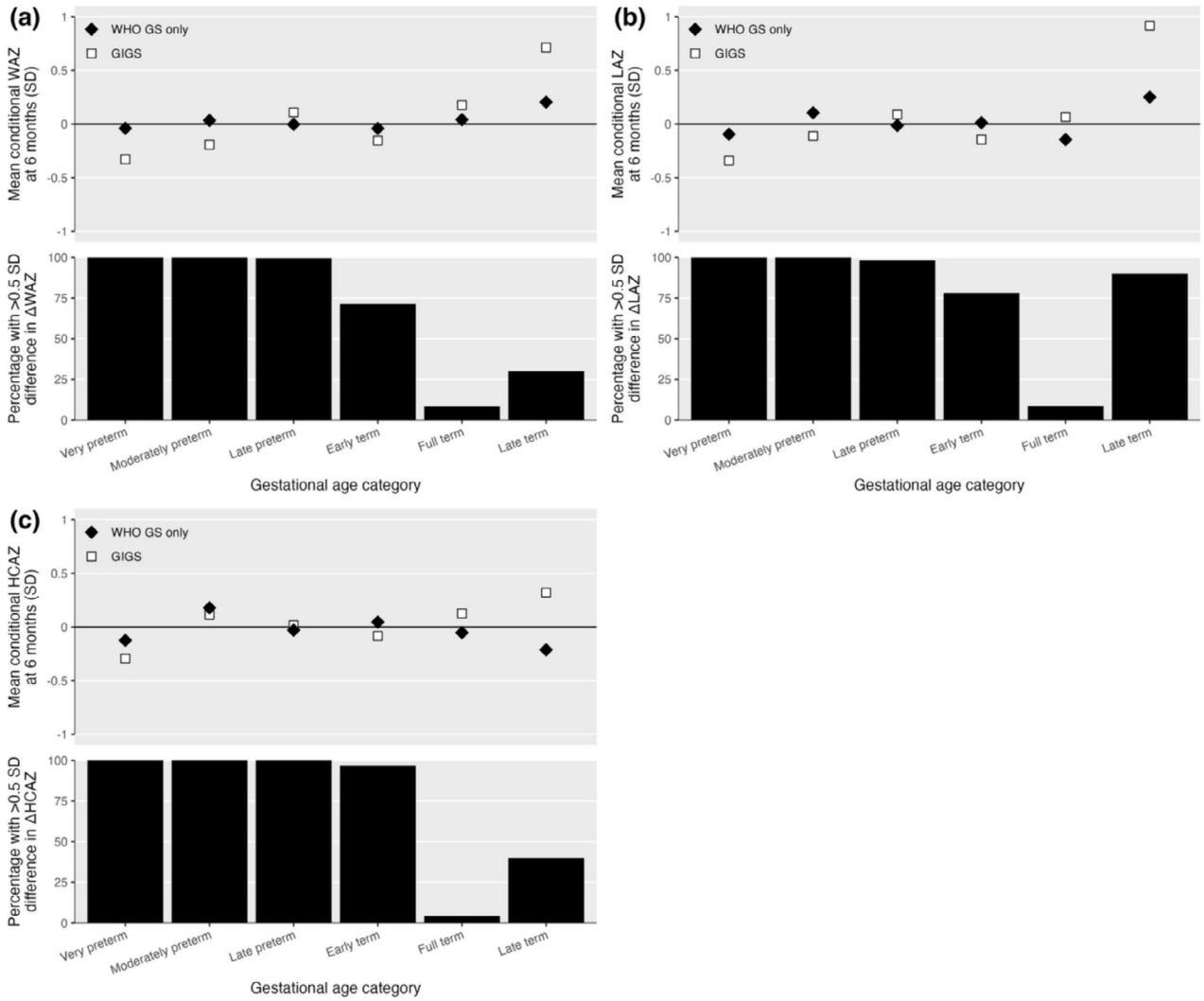

FIGURE 3 | (a–c) Mean conditional growth panels for WAZ, LAZ, HCAZ at birth to 6 months.

growth was more pronounced due to the inclusion of preterm infants, underscoring the importance of accounting for GA in growth assessment. Across postnatal intervals, especially from birth through 12 months, the GIGS approach better captured growth patterns of preterm infants, whereas both methods produced similar results in term infants.

The GIGS approach—using IG-21st NBS at birth and WHO GS postnatally for term infants, alongside IG-21st PPS—aligns more closely with biological reality. These findings support its adoption as the preferred framework for research and clinical surveillance of growth in LBW and preterm populations, where accurate classification is essential for timely intervention and consistent reporting.

5 | Conclusion

Growth monitoring in preterm infants should not rely on WHO GS alone. The GIGS-recommended approach to use IG-21st NBS and PPS in addition to WHO GS for specific infants at specific times—offers a more accurate and developmentally appropriate framework. This approach enables earlier identification of growth faltering in at-risk infants, informing timely, targeted clinical interventions and reducing the risk of under- or overtreatment. From a public health perspective, standardised adoption of GIGS tools supports more accurate reporting and tracking of child growth indicators, enhancing alignment with global nutrition and development targets. It also ensures that programmatic resources are more effectively directed to those most in need, improving the efficiency and equity of service delivery.

Author Contributions

E.O.O. and L.V. conceptualised and led the development of the GIGS project and supplement. S.P., E.O.O. and A.V. performed the analysis. E.O.O., L.V. and S.P. drafted the first version of the manuscript with further contributions from S.S. and A.V. All authors have reviewed and approved the final version.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

References

1. A. Ashworth, R. Shrimpton, and K. Jamil, "Growth Monitoring and Promotion: Review of Evidence of Impact," *Maternal & Child Nutrition* 4, no. Suppl 1 (2008): 86–117.

2. E. O. Ohuma, B. Ngatia, S. Parker, S. Strout, and L. Vesel, "Neonatal and Child Growth Monitoring: A Renewed Call for Uptake and Application of International Growth Standards," *BJOG: An International Journal of Obstetrics and Gynaecology* (2025), https://doi.org/10.1111/1471-0528.70039, Epub ahead of print.

3. B. Ngatia, L. Vesel, and E. O. Ohuma, "Growth Monitoring of Children Under Five Years Using National Reference Charts Versus International Growth Standards, 2007–2023: A Systematic Review," *BJOG: An International Journal of Obstetrics and Gynaecology* (2025), https://doi.org/10.1111/1471-0528.70068, Epub ahead of print.

4. WHO Multicentre Growth Reference Study Group, "WHO Child Growth Standards Based on Length/Height, Weight and Age," *Acta Paediatrica. Supplement* 450 (2006): 76–85, https://doi.org/10.1111/j.1651-2227.2006.tb02378.x.

5. J. Villar, D. G. Altman, M. Purwar, et al., "The Objectives, Design and Implementation of the INTERGROWTH-21st Project," *BJOG* 120, no. Suppl 2 (2013): 9–26, v, https://doi.org/10.1111/1471-0528.12047.

6. U. N. Ahmad, M. Yiwombe, P. Chisepo, T. J. Cole, G. T. Heikens, and M. Kerac, "Interpretation of World Health Organization Growth Charts for Assessing Infant Malnutrition: A Randomised Controlled Trial," *Journal of Paediatrics and Child Health* 50, no. 1 (2014): 32–39.

7. C. G. Victora, L. Adair, C. Fall, et al., "Maternal and Child Undernutrition: Consequences for Adult Health and Human Capital," *Lancet* 371, no. 9609 (2008): 340–357.

8. S. R. Parker, L. Vesel, and E. O. Ohuma, "Gigs: A Package for Standardizing Fetal, Neonatal, and Child Growth Assessment With Extensions to Egen," *Stata Journal: Promoting Communications on Statistics and Stata* 25, no. 3 (2025): 499–525, (Original work published 2025), https://doi.org/10.1177/1536867X251365448.

9. Stata github LSHTM-GIGS, *GIGS-Stata [Internet]* (GitHub Repository, 2025), https://github.com/lshtm-gigs/gigs-stata.

10. S. R. Parker, L. Vesel, and E. O. Ohuma, "Gigs: A Package for Standardising Fetal, Neonatal, and Child Growth Assessment," *Journal of Open Source Software* 10, no. 106 (2025): 7655, https://doi.org/10.21105/joss.07655.

11. ROpenSci, *gigs: Growth Standards Toolkit for R* (GitHub, 2025), https://www.github.com/lshtm-gigs/gigs/.

12. SASPAC, *gigs: Growth Standards Toolkit for SAS* (GitHub, 2025), https://github.com/SASPAC/gigs/.

13. E. O. Ohuma, S. R. Parker, S. Strout, B. Ngatia, and L. Vesel, "Scenario-Based Guidance for International Growth Standards (GIGS): For Whom, When, and How to Apply the INTERGROWTH-21st and WHO Child Growth Standards," *BJOG: An International Journal of Obstetrics and Gynaecology* (2026), https://doi.org/10.1111/1471-0528.70159, Epub ahead of print.

14. K. E. A. Semrau, R. R. Mokhtar, K. Manji, et al., "Facility-Based Care for Moderately Low Birthweight Infants in India, Malawi, and Tanzania," *PLOS Global Public Health* 3, no. 4 (2023): e0001789, https://doi.org/10.1371/journal.pgph.0001789.

15. L. Vesel, L. Spigel, J. N. Behera, et al., "Mixed-Methods, Descriptive and Observational Cohort Study Examining Feeding and Growth Patterns Among Low Birthweight Infants in India, Malawi and Tanzania: The LIFE Study Protocol," *BMJ Open* 11, no. 12 (2021): e048216, https://doi.org/10.1136/bmjopen-2020-048216.

16. L. Vesel, E. Benotti, S. Somji, et al., "Facilitators, Barriers, and Key Influencers of Breastfeeding Among Low Birthweight Infants: A Qualitative Study in India, Malawi, and Tanzania," *International Breastfeeding Journal* 18, no. 1 (2023): 59, https://doi.org/10.1186/s13006-023-00597-7.

17. L. Vesel, R. M. Bellad, K. Manji, et al., "Feeding Practices and Growth Patterns of Moderately Low Birthweight Infants in Resource-Limited Settings: Results From a Multisite, Longitudinal Observational Study," *BMJ Open* 13, no. 2 (2023): e067316, https://doi.org/10.1136/bmjopen-2022-067316.

18. F. Saidi, R. R. Mokhtar, I. F. Hoffman, et al., "Changes in Moderately Low Birthweight Infant Feeding, Care, and Health Outcomes Before Compared to During the COVID-19 Pandemic in Malawi," *Journal of Global Health* 13 (2023): 06025, https://doi.org/10.7189/jogh.13.06025.

19. T. Mvalo, S. M. Dhaded, K. P. Manji, et al., "Mortality, Morbidity and Growth Among Moderately Low Birthweight Infants in India, Malawi, and Tanzania," *BMC Pediatrics* 25, no. 1 (2025): 316, https://doi.org/10.1186/s12887-025-05668-8.

20. A. Self, M. Schlussel, G. S. Collins, et al., "External Validation of Models to Estimate Gestational Age in the Second and Third Trimester Using Ultrasound: A Prospective Multicentre Observational Study," *BJOG: An International Journal of Obstetrics and Gynaecology* 131, no. 13 (2024): 1862–1873, https://doi.org/10.1111/1471-0528.17922.

21. L. M. Dubowitz, V. Dubowitz, and C. Goldberg, "Clinical Assessment of Gestational Age in the Newborn Infant," *Journal of Pediatrics* 77 (1970): 1–10, https://doi.org/10.1016/S0022-3476(70)80038-5.

22. N. Perumal, E. O. Ohuma, A. M. Prentice, et al., "Implications for Quantifying Early Life Growth Trajectories of Term-Born Infants Using INTERGROWTH-21st Newborn Size Standards at Birth in Conjunction With World Health Organization Child Growth Standards in the Postnatal Period," *Paediatric and Perinatal Epidemiology* 36, no. 6 (2022): 839–850, https://doi.org/10.1111/ppe.12880.

23. H. Wickham, *ggplot2: Elegant Graphics for Data Analysis* (Springer-Verlag, 2016), https://doi.org/10.1007/978-3-319-24277-4_9.

24. R Core Team, *R: A Language and Environment for Statistical Computing. Version 4.3.3* (R Foundation for Statistical Computing, 2023), https://www.R-project.org/.

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25. P. Scherdel, L. Dunkel, P. van Dommelen, et al., "Growth Monitoring as an Early Detection Tool: A Systematic Review," *Lancet Diabetes and Endocrinology* 4, no. 5 (2016): 447–456, https://doi.org/10.1016/S2213-8587(15)00392-7.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Supporting Information **Figure S1:** (a–c) Proportion of infants with >0.5 SD difference in *z*-scores between WHO and GIGS standards for (a) weight-for-age (WAZ), (b) length-for-age (LAZ), and (c) head circumference-for-age (HCAZ), by gestational age category. **Figure S2:** (a–c) Birth—6 months ΔWAZ, ΔLAZ, ΔHCAZ. **Figure S3:** (a–c) 6–12 months ΔWAZ, ΔLAZ, ΔHCAZ. **Figure S4:** (a–c) 12–24 months ΔWAZ, ΔLAZ, ΔHCAZ. **Figure S5:** (a–c) Conditional growth models from birth to 6 months (WAZ, LAZ, HCAZ). **Figure S6:** (a–c) Conditional growth models from 6 to 12 months (WAZ, LAZ, HCAZ). **Figure S7:** (a–c) Conditional growth model panels from 6 to 12 months (WAZ, LAZ, HCAZ). **Figure S8:** (a–c) Conditional growth plots from 12 to 24 months (WAZ, LAZ, HCAZ). **Figure S9:** (a–c) 12–24 months conditional growth panels (WAZ, LAZ, HCAZ). **Table S1:** Baseline characteristics of moderately low birthweight infants in four sites. **Table S2:** Mean changes in WAZ, LAZ, and HCAZ *z*-scores from 6 to 12 months. **Table S3:** Mean changes in WAZ, LAZ, and HCAZ *z*-scores from 12 to 24 months. **Table S4:** Mean conditional growth in WAZ, LAZ, and HCAZ from birth to 6 months. **Table S5:** Mean conditional growth in WAZ, LAZ, and HCAZ from 6 to 12 months. **Table S6:** Mean conditional growth in WAZ, LAZ, and HCAZ from 12 to 24 months.

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